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Sub 60nm etchless MRAM devices by ion beam etching fabricated T-shaped bottom electrode

Abstract

A first conductive layer is patterned and trimmed to form a sub 30 nm conductive via on a first bottom electrode. The conductive via is encapsulated with a first dielectric layer and planarized to expose a top surface of the conductive via. A second conductive layer is deposited over the first dielectric layer and the conductive via. The second conductive layer is patterned to form a sub 60 nm second conductive layer wherein the conductive via and second conductive layer together form a T-shaped second bottom electrode. MTJ stacks are deposited on the T-shaped second bottom electrode and on the first bottom electrode wherein the MTJ stacks are discontinuous. A second dielectric layer is deposited over the MTJ stacks and planarized to expose a top surface of the MTJ stack on the T-shaped second bottom electrode. A top electrode contacts the MTJ stack on the T-shaped second bottom electrode plug.

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Background/Summary

PRIORITY DATA (1) The present application is a continuation application of U.S. application Ser. No. 17/815,971, filed Jul. 29, 2022, which is a continuation application of U.S. application Ser. No.

17/121,394, filed Dec. 14, 2020, which is a continuation application of U.S. application Ser. No. 16/452,909, filed Jun. 26, 2019, which is a divisional application of U.S. application Ser. No. 16/008,629, filed Jun. 14, 2018, each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) This application relates to the general field of magnetic tunneling junctions (MTJ) and, more particularly, to etchless methods for forming sub 60 nm MTJ structures.

BACKGROUND

(2) Fabrication of magnetoresistive random-access memory (MRAM) devices normally involves a sequence of processing steps during which many layers of metals and dielectrics are deposited and then patterned to form a magnetoresistive stack as well as electrodes for electrical connections. To define those millions of MTJ cells in each MRAM device and make them non-interacting to each other, precise patterning steps including reactive ion etching (RIE) are usually involved. During RIE, high energy ions remove materials vertically in those areas not masked by photoresist, separating one MTJ cell from another. However, the high energy ions can also react with the non-removed materials, oxygen, moisture and other chemicals laterally, causing sidewall damage and lowering device performance. To solve this issue, pure physical etching techniques such as ion beam etching (IBE) have been applied to etch the MTJ stack to avoid the damaged MTJ sidewall. However, due to their non-volatile nature, IBE etched conductive materials in the MTJ and bottom electrode can be re-deposited into the tunnel barrier, resulting in shorted devices. A new device structure and associated process flow which can form MTJ patterns with desired sizes without plasma etch is desired.

(3) Several patents teach methods of forming an MTJ without etching, including U.S. Pat. No. 9,029,170 (Li et al) and Patent CN107342331 (Wang et al), but these methods are different from the present disclosure.

SUMMARY

(4) It is an object of the present disclosure to provide a method of forming MTJ structures without chemical damage or re-deposition of metal materials on the MTJ sidewalls.

(5) Another object of the present disclosure is to provide a method of electrically isolatedly forming MTJ patterns on top of a T-shaped bottom electrode without using a plasma etch.

(6) Another object of the present disclosure is to provide a T-shaped bottom electrode and electrically isolatedly forming MTJ patterns on top of the bottom electrode without etching.

(7) In accordance with the objectives of the present disclosure, a method for fabricating a magnetic tunneling junction (MTJ) structure is achieved. A first conductive layer is deposited on a first bottom electrode. The first conductive layer is patterned and trimmed to form a sub 30 nm conductive via on the first bottom electrode. The conductive via is encapsulated with a first dielectric layer. The first dielectric layer is planarized to expose a top surface of the conductive via. A second conductive layer is deposited over the first dielectric layer and the conductive via. The second conductive layer is patterned to form a sub 60 nm second conductive layer wherein the conductive via and second conductive layer together form a T-shaped second bottom electrode. MTJ stacks are deposited on the T-shaped second bottom electrode and on the first bottom electrode wherein the MTJ stacks are discontinuous. A second dielectric layer is deposited over the MTJ stacks and planarized to expose a top surface of the MTJ stack on the T-shaped second bottom electrode. A top electrode layer is deposited on the second dielectric layer and contacting the top surface of the MTJ stack on the T-shaped second bottom electrode plug to complete the MTJ structure.

(8) Also in accordance with the objects of the present disclosure, an improved magnetic tunneling junction (MTJ) is achieved. The MTJ structure comprises a sub-60 nm MTJ device on a T-shaped

second bottom electrode, a first bottom electrode underlying the T-shaped second bottom electrode, and a top electrode overlying and contacting the MTJ device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the accompanying drawings forming a material part of this description, there is shown:

(2) FIGS. **1** through **9** illustrate in cross-sectional representation steps in a preferred embodiment of the present disclosure.

DETAILED DESCRIPTION

(3) In the present disclosure, it is demonstrated that by using a high angle ion beam etching, we can create a T shaped bottom electrode. Since the bottom portion is only sub 30 nm, much smaller than the top portion of sub 60 nm, the later MTJ deposition cannot form a continuous film along the electrode, but forms separate patterns on top. Using this etchless process, any chemical damage and/or conductive metal re-deposition on the MTJ sidewall are avoided, improving the MRAM device performance.

(4) In a typical MTJ process, the MTJ stack is deposited onto a uniformly sized bottom electrode. Plasma etch is used to transfer the photolithography created photoresist pattern into the MTJ stack. A physical etch such as pure Ar RIE or IBE can avoid chemical damage, but the metal re-deposition in this type of etch can cause electrically shorted devices. However, in the process of the present disclosure, the MTJ stack is deposited onto a T-shaped electrode, so that the patterns are formed without using plasma etch, avoiding these issues.

(5) The preferred embodiment of the present disclosure will be described in more detail with reference to FIGS. **1-9**. FIG. **1** illustrates a first bottom electrode layer **12** formed on a semiconductor substrate, not shown. The first bottom electrode **12** is preferably Ta, TaN, Ti or TiN. On top of first bottom electrode or circuit **12**, a conductive layer **14** such as Ta, TaN, Ti, TiN W, Cu, Mg, Ru, Cr, Co, Fe, Ni or their alloys is deposited to a thickness h**1** of 10-100 nm, and preferably ≥50 nm. A dielectric layer **16** such as SiO2, SiN, SiON, SiC, or SiCN is deposited using chemical vapor deposition (CVD) or spin-coating to a thickness h**2** of ≥90 nm.

(6) Next, a photoresist is spin-coated and patterned by photolithography, such as 248 nm photolithography, forming photoresist patterns **18** with size d**1** of approximately 70-80 nm and height h**3** of ≥200 nm.

(7) Now, the dielectric layer **16** and conductive layer **14** are etched by a fluorine carbon based plasma such as CF **4** or CHF **3** alone, or mixed with Ar and N.sub.2. O.sub.2 can be added to reduce the pillar size further. They can alternatively be patterned by a physical etch such as IBE. Metal layer **14** can also be patterned by a physical etch such as IBE or RIE using pure Ar plasma. Dependent on the thickness of the conductive layer **14**, the dielectric layer **16** can be partially consumed. The conductive layer's remaining thickness is still h**1** (≥50 nm) with pattern size d**2** of 15-70 nm, as shown in FIG. **2**.

(8) Next, a high angle IBE trimming **20** is applied to the conductive layer **14**. The high angle ranges from 70-90° with respect to the surface's normal line. After IBE trimming, as shown in FIG. **3**, the conductive layer pattern size decreases to d**3**, which can range from 10-30 nm, dependent on the IBE trimming conditions such as RF power (500-1000 W) and time (100-500 sec). Due to the protection of the remaining dielectric layer **16** on top of the conductive layer **14** (FIG. **2**), and the extremely low vertical etch rate (≤5 A/sec) of IBE at such a large angle, the remaining conductive layer height h**4** is the same as h**1**, or decreases less than 5 nm after this step. Ex-situ IBE trimming is used when the conductive layer **14** is made of inert metals and in-situ IBE trimming is needed for metals that can be readily oxidized in air. Compared to the immersion 193 nm or EUV photolithography which is widely used to deliver similar results in the integrated circuit (IC)

industry, this high angle IBE trimming is a much lower cost method. The remaining dielectric layer **16** and photoresist pattern **18** are consumed during the IBE trimming.

(9) As illustrated in FIG. **4**, a dielectric material **22** such as SiO.sub.2, SiN, SiON, SiC, SiCN, or amorphous carbon is deposited to a thickness of ≥100 nm to encapsulate the conductive via **14**. The dielectric material **22** may be deposited by physical vapor deposition (PVD), chemical vapor deposition (CVD), or atomic layer deposition (ALO). Whether ex-situ or in-situ PVD/CVD/ALD encapsulation is used is dependent on how sensitive these vias are to the atmosphere. Spin-on or amorphous carbon can also be used if the bottom electrode is made of an inert metal.

(10) Chemical mechanical polishing (CMP) is applied to smooth the surface as well as expose the conductive vias **14** underneath, with remaining via height h**5** of ≥45 nm, as shown in FIG. **5**. Conductive via **14** forms the bottom pillar portion of the T-shaped bottom electrode. Now, the top portion of the T-shape will be formed.

(11) A metal layer **24** such as Ta, TaN, Ti or TiN W, Cu, Mg, Ru, Cr, Co, Fe, Ni or their alloys is deposited with a thickness h**6** of 10-100 nm and preferably ≥50 nm over the via **14** and planarized dielectric layer **22**, as shown in FIG. **5**. Dielectric layer **26** such as SiO.sub.2, SiN, SiON, SiC, or SiCN with thickness h**7** of ≥20 nm is deposited over metal layer **24**. Photoresist with a thickness h**8** of ≥200 nm is deposited and patterned by 248 nm photolithography to form photoresist mask **28**. The dielectric **26** and metal layer **24** are etched by RIE, IBE or their combination to form the pattern size d**4** of 50-60 nm, as shown in FIG. **6**.

(12) FIG. **6** illustrates the T-shaped bottom electrode **14**/**24** of the present disclosure. The dielectric encapsulation **22** and remaining hard mask **26** are stripped off to expose the entire T-shaped electrode. A fluorine carbon plasma with high carbon/fluorine ratio such as C.sub.4F.sub.8 or CH.sub.2F.sub.2 can be used to strip off materials like SiO.sub.2, SiN, SiON, SiC, or SiCN, without etching the T-shaped bottom electrode. O.sub.2 plasma can be used to strip off spin-on or CVD deposited carbon encapsulation. Ex-situ stripping is used when the metal vias are made of inert metals, but in-situ stripping is needed for metals that can be readily oxidized in air.

(13) Now, as shown in FIG. **7**, MTJ film layers are deposited, typically including a seed layer, a pinned layer, a barrier layer, a free layer, and a cap layer, for example. These layers form the MTJ film stack **30**. The MTJ stack **30** can be deposited ex-situ, but preferrably, the MTJ stack is deposited in-situ. After the MTJ stack is deposited, it only covers the top of the T-shaped bottom electrode **14**/**24** as well as the first bottom electrode **12** on the sides. It should be noted that the MTJ stack is discontinuous because of the undercut structure **24**/**14**.

(14) As a result, the MTJ patterns with size d**5** (50-60 nm) are formed without plasma etch and thus, without plasma etch-induced chemical damage and/or conductive metal re-deposition on the MTJ sidewalls. Now, as shown in FIG. **8**, dielectric layer **32** is deposited and flattened by CMP, for example, wherein the top MTJ surface is exposed. Finally, the top metal electrode **34** is deposited to form the whole device, also preferably in an in-situ method, as shown in FIG. **9**.

(15) In the process of the present disclosure, by decoupling the etch process, we can use a high angle ion beam etching to create a T-shaped bottom electrode to allow for etchless MTJ patterns. The top and pillar T-shaped electrode portions' sizes are sub 60 nm and 30 nm, respectively. After MTJ deposition, the same size of 60 nm MTJ patterns can be electrically isolatedly formed on top of the bottom electrode, without using an etching process. This approach avoids any chemical damage and/or conductive metal re-deposition on the MTJ sidewall, thus improving the MRAM device performance.

(16) FIG. **9** illustrates the completed MTJ structure of the present disclosure. We used high angle IBE trimming to fabricate the T-shaped bottom electrode **14**/**24** to create MTJ patterns **30** without using a plasma etch. This approach avoids any chemical damage and/or conductive metal re-deposition on the MTJ sidewall, improving the MRAM device performance. Dielectric layer **32** covers the MTJ structures. Top electrode **34** contacted the MTJ structure **30**.

(17) The process of the present disclosure will be used for MRAM chips of size smaller than 60 nm

as problems associated with chemically damaged sidewalls and re-deposition from the bottom electrode become very severe for these smaller sized MRAM chips.
(18) Although the preferred embodiment of the present disclosure has been illustrated, and that form has been described in detail, it will be readily understood by those skilled in the art that various modifications may be made therein without departing from the spirit of the disclosure or from the scope of the appended claims.

Claims

1. A device comprising: a first electrode; a conductive feature disposed directly on the first electrode; a first magnetic tunneling junction (MTJ) structure disposed directly on the conductive feature, wherein the first MTJ structure has a top surface facing away from the first electrode; a second MTJ structure disposed directly on the first electrode, wherein the second MTJ structure has a top surface facing away from the first electrode; a dielectric layer extending from the second MTJ structure to the first MTJ structure, wherein the dielectric layer covers the top surface of the second MTJ structure; and a second electrode disposed directly on the top surface of the first MTJ.
2. The device of claim 1, further comprising a third MTJ structure disposed directly on the first electrode, wherein the third MTJ structure has a top surface facing away from the first electrode, and wherein the dielectric layer extends from the third MTJ structure to the first MTJ structure, wherein the dielectric layer covers the top surface of the third MTJ structure.
3. The device of claim 1, wherein the first MTJ structure is wider than the second MTJ structure.
4. The device of claim 1, wherein the conductive feature is formed of a first portion interfacing with the first electrode and a second portion interfacing with the second electrode, and wherein the first portion of the conductive feature has a different width than the second portion of the conductive feature.
5. The device of claim 4, wherein the first portion of the conductive feature has a different material composition than the second portion of the conductive feature.
6. The device of claim 4, wherein the first portion of the conductive feature has the same material composition as the second portion of the conductive feature.
7. The device of claim 1, wherein the first MTJ structure and the conductive feature have substantially the same width at interface between the first MTJ structure and the conductive feature.
8. A device comprising: a first electrode having a first cross-sectional shape; a second electrode disposed directly on the first electrode, the second electrode having a second cross-section shape that is different than the first cross-sectional shape; a first magnetic tunneling junction (MTJ) structure disposed directly on the second electrode, wherein no portion of the first MTJ structure is disposed directly on the first electrode; a second MTJ structure disposed directly on the first electrode, wherein no portion of the second MTJ structure is disposed directly on the second electrode; a dielectric layer extending from the second MTJ structure to the first MTJ structure, the dielectric layer being formed of the same material throughout the dielectric layer, wherein a top surface of the first MTJ structure and a top surface of the dielectric layer are coplanar, wherein the dielectric layer extends from a top surface of the second MTJ structure to the third electrode, and wherein the top surface of the second MTJ structure faces away from the first electrode.
9. The device of claim 8, wherein the dielectric layer further extends from the second electrode to the first electrode.
10. The device of claim 8, wherein the dielectric layer extends from a sidewall of the first MTJ structure to a sidewall of the second MTJ structure.
11. The device of claim 8, further comprising a third electrode, wherein the first MTJ structure interfaces with the third electrode while the dielectric layer prevents the second MTJ structure from interfacing with the third electrode.

12. The device of claim 8, wherein the dielectric layer continuously extends from the top surface of the second MTJ structure to a bottom surface of the third electrode.
13. The device of claim 8, wherein the second cross-sectional shape is a T-shaped cross-sectional shape.
14. The device of claim 8, wherein a portion of a top surface of the first electrode extends from the second electrode to the second MTJ structure, and wherein the dielectric layer interfaces with and covers the portion of the top surface of the first electrode.
15. The device of claim 8, wherein the first MTJ structure and the second MTJ structure are formed of the same material layers.
16. A device comprising: a first electrode; a T-shaped electrode disposed directly on the first electrode; a first magnetic tunneling junction (MTJ) structure disposed directly on the T-shaped electrode and extending to a first height above the first electrode; a second MTJ structure disposed directly on the first electrode and extending to a second height above the first electrode that is less than the first height, wherein the first MTJ structure and the second MTJ structure are formed of the same material layers; and an encapsulation layer extending from the second MTJ structure to the first MTJ structure, the encapsulation layer being formed of the same material throughout the encapsulation layer, wherein the encapsulation layer covers a top surface of the second MTJ structure.
17. The device of claim 16, further comprising a third MTJ structure disposed directly on the first electrode and extending to a third height over the first electrode, the third height being different than the first height, and wherein the second MTJ structure is disposed on a first side of the T-shaped electrode and the third MTJ structure is disposed on a second side of the T-shaped electrode, the second side of the T-shaped electrode being opposite the first side of the T-shaped electrode.
18. The device of claim 17, wherein the third height is substantially the same as the second height, and wherein the encapsulation layer extends from the third MTJ structure to the first MTJ structure.
19. The device of claim 16, wherein the T-shaped electrode includes a material selected from the group consisting of Ta, TaN, Ti, TiN, W, Cu, Mg, Ru, Cr, Co, Fe and Ni.
20. The device of claim 16, wherein the T-shaped electrode includes a lower portion formed of a first material and an upper portion formed of a second material that is different than the first material.